Comparative Assessment: DRBENCHMARK vs. BLURB

Deployment context: EU Pharmaceutical Regulatory NLP — French

Deployment Context

Use case: In a pharmaceutical setting, a regulatory specialist uses document management software to ensure compliance in French drug labeling and scientific abstracts by applying a model evaluated on semantic textual similarity and fine-grained named-entity recognition. The system identifies over a dozen types of precise entities, such as chemical compounds and dosage modes, while verifying that safety warnings across different leaflets or patents are semantically consistent. These results determine whether pharmacological documentation meets strict professional standards for official submission or requires manual revision to prevent regulatory non-compliance.

Target population: Regulatory affairs specialists, pharmacologists, and legal experts at pharmaceutical companies or government health agencies.

Overall Risk Ratings

Benchmark	Risk	Avg. Score
DRBENCHMARK	HIGH	2.5 / 5
BLURB	HIGH	1.8 / 5

Dimension Scores Comparison

Dimension	DRBENCHMARK	Rating	BLURB	Rating	Δ
Input Ontology (IO)	2/5	Significant gaps	2/5	Significant gaps	0
Input Content (IC)	2/5	Significant gaps	1/5	Serious concern	+1
Input Form (IF)	4/5	Minor gaps	1/5	Serious concern	+3
Output Ontology (OO)	1/5	Serious concern	2/5	Significant gaps	-1
Output Content (OC)	2/5	Significant gaps	2/5	Significant gaps	0
Output Form (OF)	4/5	Minor gaps	3/5	Moderate gaps	+1
Average	**2.5**		**1.8**		**+0.7**

Δ = DRBENCHMARK score − BLURB score. Positive = regional benchmark scores higher.

Overall Summaries

DRBENCHMARK (risk: HIGH)

DrBenchmark is the most comprehensive French biomedical NLU benchmark currently available [Q16, Q83], and parts of its content (QUAERO EMEA, Mantra-GSC fr_emea, DEFT2020 drug-leaflet pairs, PxCorpus posology schema) provide partial signal for the EU pharmaceutical regulatory deployment. However, the deployment's HIGH-priority dimensions — input ontology, input content, output ontology, and output content — show severe misalignment. The benchmark is calibrated for clinical and research French biomedical NLP, not for EMA/ANSM regulatory affairs workflows: (i) regulatory document genres (SmPCs, CTD modules, EU-RMPs) are absent; (ii) regulatory-specific entity types (INN as distinct from excipient, ATC codes, EMA-templated posology, contraindication qualifiers) are not annotated as distinct classes; (iii) STS scoring is calibrated for general semantic proximity and demonstrably tolerates legally significant micro-differences (2×–60× numeric divergences score 3.0–4.0); (iv) annotators are biomedical NLP / clinical experts, not regulatory affairs specialists, and elicitation explicitly anticipates systematic disagreement on borderline cases. Lower-priority dimensions (input form, output form) are well aligned. Web search confirms several of these gaps are field-level, not just benchmark-level [WEB-11, WEB-12, WEB-16].

BLURB (risk: HIGH)

BLURB is a well-constructed benchmark for English-language PubMed-based biomedical NLP, but it is fundamentally misaligned with the EU/French pharmaceutical regulatory NLP deployment across five of six validity dimensions. The language gap (English-only [Q1, Q126] vs French-language deployment), document-genre gap (PubMed abstracts vs SmPCs/PILs/CTD modules), entity-ontology gap (no INN/ATC/posology/excipient label spaces), STS-construct gap (general biomedical proximity [Q74] vs legally-determinative regulatory equivalence), and annotator-norm gap (biomedical researchers vs EMA/ANSM-aligned regulatory experts) are each rated HIGH severity by the elicitation and confirmed by web research [WEB-9, WEB-10, WEB-12, WEB-23]. Output form is the only dimension with reasonable representational alignment, and even there, the absence of calibration/threshold-sensitivity reporting limits direct utility for a human-review-triggering compliance pipeline. No remediation can be done within BLURB itself; bespoke French regulatory benchmark development against the EMA AI Reflection Paper [WEB-6] normative framework is required.

Per-Dimension Justification Comparison

Each benchmark's full assessment justification and cited evidence, shown side by side.

Input Ontology (IO)

Whether the benchmark's test case categories cover the query types expected in deployment.

DRBENCHMARK — 2/5 (Significant gaps) [high confidence]	BLURB — 2/5 (Significant gaps) [high confidence]
DrBenchmark's task taxonomy (POS, NER, MCQA, MCC, MLC, STS) is calibrated for clinical and research NLP rather than EU pharmaceutical regulatory workflow categories [Q81, Q83]. The deployment requires NER and STS over EU regulatory document genres (SmPCs, PILs, CTD modules, EU-RMPs), but the benchmark organizes tasks around general biomedical constructs and includes substantial irrelevant categories: MCQA over pharmacy exam questions [Q39], abstract-level specialty classification [Q38], ICD-10 chapter classification [Q45], and negation/speculation scope [CAS, ESSAI]. The dataset analysis confirms critical task-type mismatches for FrenchMedMCQA, MORFITT, and DiaMED, which evaluate constructs orthogonal to the deployment's NER/STS extraction needs (FRENCHMEDMCQA-D11, MORFITT-D7, DIAMED-D6, DIAMED-D13). No category covers regulatory equivalence scoring as a distinct output decision rule, despite this being a HIGH-priority deployment dimension. Partial overlap exists for NER (QUAERO, MANTRAGSC, DEFT2021, PxCorpus) and STS (CLISTER, DEFT2020), so the benchmark is not entirely orthogonal — but construct underrepresentation for regulatory-specific task types is severe.	BLURB's task taxonomy (NER, PICO, relation extraction, sentence similarity, document classification, QA) is explicitly scoped to PubMed-based biomedical applications [Q46, Q48], with clinical and 'other high-value verticals' deferred to future work [Q46, Q197]. The deployment requires categories for regulatory document genres (SmPCs, PILs, CTD modules, patent claims) and regulatory-specific tasks such as cross-document consistency checking and regulatory equivalence STS — none of which are represented in BLURB's taxonomy. The QA scope is further restricted to yes/no and three-way classification [Q49, Q51, Q124], excluding the kind of structured field-extraction or compliance-verification task structure relevant to regulatory submissions. Web-confirmed: no public regulatory NLP benchmark with these task categories exists for French [WEB-12, WEB-23], reinforcing the structural gap. The taxonomy does include NER, document classification, and STS at an abstract level, which are the surface task types the deployment also uses, so the gap is one of category specificity rather than total absence.

DRBENCHMARK evidence	BLURB evidence
[Q81] 'Table 8 summarizes the results obtained on average by the considered MLMs when aggregating the tasks into one of the five designated categories: POS, NER, MCQA, MCC (Multi-class classification), MLC (Multi-label classification), or STS tasks.' (p.8)	[Q46] 'We focus on PubMed-based biomedical applications, and leave the exploration of the clinical domain, and other high-value verticals to future work.' (p.6)

[Q83] 'In this paper, we introduced DrBenchmark, the first large language understanding benchmark tailored for the French biomedical domain.' (p.9)	[Q48] 'BLURB is comprised of a comprehensive set of biomedical NLP tasks from publicly available datasets, including named entity recognition (NER), evidence-based medical information extraction (PICO), relation extraction, sentence similarity, document classification, and question answering.' (p.7)
[Q39] 'FrenchMedMCQA ... contains 3,105 questions coming from real exams of the French medical specialization diploma in pharmacy' (p.4)	[Q49] 'For question answering, prior work has considered both classification tasks ... and more complex tasks such as list and summary' (p.7)
	[Q51] 'For simplicity, we focus on the classification tasks such as yes/no question-answering in BLURB' (p.7)
	[Q197] 'Future directions include: ... extension of the BLURB benchmark to clinical and other high-value domains.' (p.22)
	[WEB-12] DART (Italian SmPC NER/RE corpus, 2025) is the closest published regulatory-document NLP resource; no French equivalent exists
	[WEB-23] Practitioner review confirms NLP for EU SmPC/prescribing information remains hybrid with no validated regulatory-specific benchmark

Input Content (IC)

Whether individual datapoint content is culturally and contextually appropriate for the target region.

DRBENCHMARK — 2/5 (Significant gaps) [high confidence]	BLURB — 1/5 (Serious concern) [high confidence]
Input content overlaps only narrowly with the regulatory deployment. QUAERO's EMEA subset and Mantra-GSC's EMEA subset are the only two corpora containing genuine EMA-sourced text [Q33, Q34, Q40], and the dataset analysis confirms these are PIL-register drug leaflets rather than SmPC text, with QUAERO EMEA holding only ~10 segmented documents across train/dev/test (QUAERO-D1, QUAERO-D9, MANTRAGSC-D11, MANTRAGSC-D12) — corroborated by [WEB-3, WEB-4]. The remaining majority of content is clinical case reports, research abstracts, clinical trial protocols, and spoken prescriptions [Q19, Q24, Q29, Q38, Q43, Q44, Q45], which differ substantially from EMA QRD-templated regulatory prose in register and formulaicity [WEB-2, WEB-7]. While pharmaceutical vocabulary (INNs, dosages, routes) does appear across corpora (DEFT2021-D1, PXCORPUS-D1, ESSAI-D1, DEFT2020-D1), the deployment's target entity nomenclature (INN/ATC/excipient distinctions, EMA-templated posology, contraindication qualifiers) is not realized in regulatory register at scale. Off-domain content (beekeeping, sports, history in DEFT2020; veterinary and non-EU geographies in MORFITT) further dilutes the relevant signal (DEFT2020-D13, DEFT2020-D14, MORFITT-D17, MORFITT-D18). Construct-irrelevant variance is therefore substantial, though not total.	BLURB's input content is drawn entirely from PubMed abstracts [Q1, Q53, Q54, Q56, Q57, Q58, Q59, Q64, Q70, Q126, Q127], with the pretraining corpus comprising 14 million abstracts [Q126]. The deployment targets French-language SmPCs, PILs, CTD modules, and patent claims — formulaic, template-governed regulatory documents that are categorically distinct from PubMed research prose. BLURB explicitly excludes even clinical notes on the grounds that they 'differ substantially from biomedical literature' [Q43]; this same logic applies a fortiori to regulatory documents, which differ from research abstracts in genre, register, legal function, and language. No French-language source, no SmPC, no PIL, and no EU regulatory submission document appears anywhere in BLURB's data. Web-confirmed: only the small QUAERO EMEA subcorpus (4 documents) provides any French regulatory NLP content [WEB-19, WEB-20], and even there the entity taxonomy is research-biomedical. The 2025 Frontiers SmPC IDMP study confirms empirically that even current LLMs miss ATC codes and excipient dosages in SmPC extraction [WEB-9], evidencing the magnitude of the content gap.

DRBENCHMARK evidence	BLURB evidence

[Q19] 'A diverse range of data origins: Scientific literature, clinical trials, clinical cases, speech transcriptions, and more as described in Table 2.' (p.2)	[Q1] 'we compile a comprehensive biomedical NLP benchmark from publicly-available datasets.' (p.1)
[Q34] 'The dataset covers two text genres (drug leaflets and biomedical titles), consisting of a total of 103,056 words sourced from EMEA or MEDLINE.' (p.3)	[Q43] 'Clinical notes differ substantially from biomedical literature, to the extent that we observe BERT models pretrained on clinical notes perform poorly on biomedical tasks' (p.6)
[Q40] 'Mantra-GSC ... 3 sources ... biomedical abstracts/titles, drug labels and patents' (p.4)	[Q126] 'For biomedical domain-specific pretraining, we generate the vocabulary and conduct pretraining using the latest collection of PubMed abstracts: 14 million abstracts, 3.2 billion words, 21 GB.' (p.12)
[Q44] 'PxCorpus ... 4 hours (1,981 recordings) of transcribed and annotated dialogues focused on drug prescriptions.' (p.4)	[WEB-9] 2025 Frontiers in Medicine SmPC IDMP study — ATC codes and excipient dosages frequently missed by LLMs, confirming SmPC content is empirically distinct from biomedical research prose
[Q24] 'DEFT-2021 ... is a subset of 275 clinical cases' (p.3)	[WEB-19] QUAERO French Medical Corpus — only public French regulatory-document NLP resource, very small EMEA subcorpus
[WEB-3] QUAERO EMEA HuggingFace dataset card confirms PIL leaflet text, not SmPC text	[WEB-20] QUAERO HuggingFace dataset documentation — 4 EMEA documents segmented into 15 sub-documents
[WEB-4] QUAERO official corpus page documents EMEA/MEDLINE source genres	
[WEB-7] EMA QRD template v10.4 mandates standard statements 'must be used whenever applicable' — register absent from benchmark at scale	

Input Form (IF)

Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

DRBENCHMARK — 4/5 (Minor gaps) [high confidence]	BLURB — 1/5 (Serious concern) [high confidence]
Input form is largely well-aligned. Both deployment and benchmark operate on French text in Latin script with standard diacritics, and the benchmark filters multilingual datasets to French-only [Q31]. No script, modality, or infrastructure mismatch exists for the core deployment. Two concerns reduce the score from 5: (1) sentence-splitting of long EMEA documents [Q37] introduces structural artifacts (QUAERO-D19, QUAERO-D20) and removes document-level context that SmPC/CTD reasoning may require; and (2) PxCorpus introduces transcribed speech with ASR artifacts (PXCORPUS-D10, PXCORPUS-D12, PXCORPUS-D13) that are absent from the deployment's text-only document-management input modality. The benchmark also exposes non-French configs in E3C and MANTRAGSC HuggingFace repositories (E3C-D8, E3C-D9) that could contaminate evaluation if pipelines do not filter correctly. Tokenization variation is studied [Q78–Q80] and shown to be a minor effect.	BLURB is exclusively English-language text [Q1, Q46, Q126], with vocabulary generated from PubMed [Q126] and tokenization optimized for English biomedical terminology [Q29, Q162, Q168]. The deployment operates on French-language regulatory documents with French diacritics and French-specific tokenization needs. No multilingual processing, no French-language variant, and no porting strategy is documented in BLURB. The benchmark's own argument that vocabulary mismatch causes performance degradation — 'standard BERT models are forced to divert parametrization capacity and training bandwidth to model biomedical terms using fragmented subwords' [Q28], with examples like 'naloxone' fragmenting into four pieces [Q29] — applies with even greater force across the English↔French boundary. Sequence-length truncations (128/256/512 tokens [Q120, Q125]) were calibrated for PubMed sentence/abstract structure, not for SmPC sections or PIL prose. Web-confirmed French biomedical models (DrBERT, CamemBERT-bio, AliBERT) [WEB-14, WEB-15, WEB-16, WEB-17] exist but are not part of BLURB and have not been fine-tuned on EMA/ANSM regulatory genres. Form mismatch is foundational.

DRBENCHMARK evidence	BLURB evidence
[Q31] 'While the dataset covers 5 languages, only the French portion is retained for the benchmark.' (p.3)	[Q1] 'we compile a comprehensive biomedical NLP benchmark from publicly-available datasets.' (p.1)
[Q37] 'considering that some documents from EMEA exceed the maximum input sequence length that most current language models can handle, we decided to split these documents into sentences.' (p.3)	[Q28] 'standard BERT models are forced to divert parametrization capacity and training bandwidth to model biomedical terms using fragmented subwords.' (p.5)
[Q78] 'FlauBERT is the model with the least word segmentation (1.43 in average), while DrBERT-CP tends to have the highest average segmentation (1.90 in average).' (p.8)	[Q29] 'naloxone, a common medical term, is divided into four pieces ... acetyltransferase is shattered into seven pieces' (p.5)
[Q80] 'tokenization, as it is currently done in MLMs, seems to play a minor role in the performance of systems.' (p.8)	[Q120] 'we use padding or truncation to set the input length to 128 tokens for GAD and 256 tokens for ChemProt and DDI' (p.12)
	[Q126] 'we generate the vocabulary and conduct pretraining using the latest collection of PubMed abstracts: 14 million abstracts, 3.2 billion words' (p.12)
	[WEB-14] DrBERT — French biomedical PLM pretrained from scratch (ACL 2023), not part of BLURB
	[WEB-15] CamemBERT-bio — French biomedical continual-pretraining model (LREC-COLING 2024)
	[WEB-16] AliBERT — French biomedical PLM with Unigram tokenizer (BioNLP 2023)

	[WEB-17] DrBenchmark — 20-task French biomedical evaluation suite, separate from BLURB and not regulatory-specific

Output Ontology (OO)

Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

DRBENCHMARK — 1/5 (Serious concern) [high confidence]	BLURB — 2/5 (Significant gaps) [high confidence]
Output ontology shows fundamental misalignment with the deployment, which the elicitation marked HIGH priority. Two critical violations are present. (1) STS scoring uses general semantic proximity on a 0–5 scale [Q41, Q54], whereas the deployment requires regulatory-equivalence scoring sensitive to dose-threshold and population-qualifier micro-differences (elicitation A3). The dataset analysis empirically demonstrates the calibration is incompatible: a 2× quantitative difference scores 4.0, a 14× difference scores 3.75, and a 60× PSA difference scores 3.0 (CLISTER-D12, CLISTER-D14, CLISTER-D13); legally significant additions of specific opioid names or 'sous stricte surveillance médicale' qualifiers are not penalized in DEFT2020 (DEFT2020-D16, DEFT2020-D17, DEFT2020-D20). (2) NER label categories are calibrated to clinical/UMLS taxonomies that conflate distinct regulatory entity types: QUAERO and Mantra-GSC's CHEM tag merges INN, brand name, and excipient (QUAERO-D10, QUAERO-D11, MANTRAGSC-D5, MANTRAGSC-D14); ATC codes and EMA administrative identifiers receive O tag (QUAERO-D12); contraindication qualifiers are tagged by entity type rather than as regulatory contraindication structure (QUAERO-D16, MANTRAGSC-D4). Web search confirms no French NLP benchmark with INN/ATC/excipient/contraindication-specific schemas exists [WEB-11, WEB-12]. The leading model also does not excel at MLC or STS [Q82] — the dimensions on which the deployment most depends. This is a structural validity violation.	BLURB's output ontology consists of task-specific label spaces: BIO/BIOUL token tags for NER over chemical/disease/gene/molecular-biology entity types [Q107, Q108, Q109], five-way ChemProt relation classes [Q65], yes/no and yes/maybe/no QA labels [Q86, Q90], binary cancer hallmark indicators [Q78, Q80], and a 0–4 BIOSSES regression score [Q74, Q75]. None of these label spaces accommodates regulatory-specific entity types (INNs, ATC codes, excipient nomenclature, posology expressions, EMA-template contraindication qualifiers) or regulatory-equivalence scoring. The BIOSSES regression operationalizes general biomedical semantic proximity ['estimated similarity score in the range from 0 (no relation) to 4 (equivalent meanings)' Q74] — not the legally determinative dose-threshold/population-qualifier sensitivity the deployment requires. Web-confirmed: no regulatory-equivalence STS benchmark exists [WEB-10], and 2025 SmPC LLM extraction shows ATC codes and excipient dosages are frequently misclassified [WEB-9]. The macro-averaged BLURB summary score [Q52, Q150] does not encode any decision-boundary or human-review-trigger semantics. Some weak structural overlap exists (binary classification, regression scoring) at the abstract level, hence score 2 rather than 1.

DRBENCHMARK evidence	BLURB evidence
[Q41] 'similarity scores ranging from 0 to 5 to each pair. The scores were then averaged together to obtain a floating-point number' (p.4)	[Q65] 'we follow the ChemProt authors' suggestions and focus on classifying five high-level interactions' (p.8)
[Q54] 'For STS tasks, the models' performance was assessed using ... Spearman correlation, and ... mean relative solution distance accuracy (EDRM)' (p.6)	[Q74] 'The Sentence Similarity Estimation System for the Biomedical Domain ... 100 pairs of PubMed sentences each of which is annotated by five expert-level annotators with an estimated similarity score in the range from 0 (no relation) to 4 (equivalent meanings).' (p.9)
[Q82] 'the leading model, DrBERT-FS, does not excel in tasks such as MLC or STS.' (p.8)	[Q75] 'It is a regression task, with the average score as the final annotation.' (p.9)
[WEB-11] Kadi et al. 2025 — ATC codes and excipient dosages 'frequently missing or misclassified' in LLM extraction from EMA SmPCs, confirming regulatory entity extraction is unsolved	[Q78] 'It contains annotation on PubMed abstracts with binary labels each of which signifies the discussion of a specific cancer hallmark.' (p.9)

[WEB-12] Bayer et al. 2021 — general drug-label NLP systems 'would need to be modified or reconfigured to lower error rates to support their use in a regulatory setting'	[Q86] 'an annotated label of whether the text contains the answer to the research question (yes/maybe/no)' (p.9)
	[Q108] 'The NER tasks in BLURB are only concerned about one entity type (in JNLPBA, all the types are merged into one).' (p.11)
	[WEB-10] EMA 2024 AI Observatory Report — internal regulatory NLP tools (Scientific Explorer, PICROSS, EurEKA, AERGIA) but no public regulatory-equivalence benchmark
	[WEB-9] 2025 Frontiers SmPC IDMP study — ATC codes and excipient dosages 'frequently missing or misclassified' by LLMs

Output Content (OC)

Whether ground-truth labels align with the judgments of regional stakeholders.

DRBENCHMARK — 2/5 (Significant gaps) [high confidence]	BLURB — 2/5 (Significant gaps) [high confidence]
Annotator alignment with regulatory standards is largely undocumented and where documented, is inconsistent with the deployment's ground-truth norms. Annotation provenance is documented for only three datasets — DEFT-2021 (manual NER and MLC) [Q25], CLISTER (multiple annotators averaged) [Q41], DiaMED (several annotators including one medical expert, with Cohen's Kappa and Gwet's AC1) [Q45, Q46], and CAS POS (automatic via Tagex 3) [Q42]. None of these involve regulatory affairs or pharmacovigilance specialists; benchmark author affiliations are academic NLP and clinical medicine [Q8]. The elicitation (A4) confirms systematic disagreement is anticipated on borderline cases. Empirical dataset evidence corroborates: CLISTER scoring patterns are inconsistent with regulatory-equivalence norms (CLISTER-D12, CLISTER-D17, CLISTER-D18); DEFT2020 annotators score legally significant qualifier additions/deletions as near-equivalent (DEFT2020-D16, DEFT2020-D17, DEFT2020-D20); QUAERO/Mantra-GSC clinical UMLS conventions tag pregnancy contraindications by entity type rather than regulatory construct (QUAERO-D16, MANTRAGSC-D4). Convergent and external validity are both threatened. The score is 2 rather than 1 because some annotation processes are documented with multi-annotator IAA (DiaMED) and CLISTER's exact-paraphrase ceiling behavior shows annotator reliability on unambiguous cases (DEFT2020-D8).	BLURB's annotator population is biomedical-research-oriented and heterogeneous: Amazon Mechanical Turkers and Upwork contributors with medical training for EBM PICO [Q60], 'five expert-level annotators' for BIOSSES [Q74], 'biomedical experts' for BioASQ [Q88], and unspecified/varied for the remaining datasets. None are regulatory affairs specialists or legal experts operating under EMA/ANSM normative frameworks. The deployment explicitly anticipates that biomedical NLP annotators will 'prioritize clinical relevance' while regulatory ground truth requires 'rigid legal constraint satisfaction,' producing systematic disagreements on borderline cases (elicitation A4). Web-confirmed: no published IAA study comparing biomedical vs regulatory annotators exists, and no regulatory-expert-annotated French pharmaceutical NER/STS dataset has been identified [WEB-12]. BLURB also does not systematically report IAA, annotator demographics, or annotation protocols across datasets. ChemProt label expansion via assigning false labels to unannotated pairs [Q66] further reflects research-pragmatic conventions that may be inappropriate where unannotated content has legal-evidentiary weight. Some annotation quality signals (BIOSSES expert annotators, EBM PICO test set quality stratification) [Q60, Q74] exist, hence score 2 rather than 1.

DRBENCHMARK evidence	BLURB evidence
[Q8] 'LIA, Avignon Université, Zenidoc, Nantes Université, CHU Nantes, Clinique des données, INSERM, CIC 1413, École Centrale Nantes, CNRS, LS2N, ... Université de Lille' (p.1)	[Q60] 'The training and development sets were labeled by Amazon Mechanical Turkers, whereas the test set was labeled by Upwork contributors with prior medical training.' (p.8)

[Q25] 'This dataset is manually annotated in two tasks: (i) multi-label classification and (ii) NER.' (p.3)	[Q66] 'we expand the training and development sets by assigning a false label for all chemical-protein pairs that occur in a training or development sentence, but do not have an explicit label in the ChemProt corpus.' (p.8)
[Q41] 'manually annotated by several annotators, who assigned similarity scores ranging from 0 to 5 to each pair. The scores were then averaged together' (p.4)	[Q74] '100 pairs of PubMed sentences each of which is annotated by five expert-level annotators' (p.9)
[Q42] 'CAS ... POS tagging with 31 classes using automatic annotations through Tagex 3 ... 98% precision.' (p.4)	[Q88] 'The BioASQ corpus contains multiple question answering tasks annotated by biomedical experts' (p.9)
[Q45] 'manually annotated by several annotators, one of which is a medical expert, into 22 chapters of the International Classification of Diseases, 10th Revision (ICD-10)' (p.4)	[WEB-12] No regulatory-expert-annotated pharmaceutical NLP dataset (French or English) identified; DART Italian corpus does not use regulatory-expert annotators
[Q46] 'Inter-annotator agreement statistics. κ is referring to Kappa Cohen and G to Gwet's AC1.' (p.4)	

Output Form (OF)

Whether the expected output modality matches regional deployment needs and accessibility.

DRBENCHMARK — 4/5 (Minor gaps) [high confidence]	BLURB — 3/5 (Moderate gaps) [medium confidence]
Output form is well-aligned. The deployment consumes label and score outputs in modalities the benchmark produces: IOB2 sequence labels via SeqEval [Q52], multiple-choice selections, multi-label classification labels, and continuous similarity scores [Q54]. Models predict only the first token label per word for tokenizer-agnostic evaluation [Q53]. Hyperparameters are documented [Q49, Q99] and best models saved on validation metrics [Q100]; results averaged over four runs with Student's t-test [Q51]. No output representation mismatch exists. Two minor concerns: (1) the EDRM metric [Q54] is calibrated for DEFT-2020 task design and has no documented regulatory calibration — this is a calibration concern that cuts across to OO; (2) compute requirements (~2,500 GPU hours [Q93], 36.9 kgCO2eq [Q94]) raise reproducibility concerns for resource-constrained users [Q92], though HuggingFace toolkit and documented hyperparameters [Q49] mitigate this. The output-form modality match is strong, justifying score 4 rather than 5.	BLURB's output forms — categorical labels for classification, BIO/BIOUL sequence tags for NER, regression scores for STS, accuracy for QA, F1 variants for classification/extraction, Pearson correlation for STS [Q76, Q83, Q100, Q176] — broadly match the representational forms the deployment requires (entity tags, similarity/equivalence scores, classification labels). At this representational level the form mismatch is modest, and the deployment elicitation rates OF as MODERATE priority. However, BLURB reports only aggregate metrics: macro-averaged BLURB score [Q52, Q150], task-level F1/accuracy/Pearson, and averages across multiple runs for variance management [Q140]. There are no confidence-calibration metrics, threshold-sensitivity analyses, or human-review-triggering decision boundaries documented [Q150, Q151, Q152] — precisely the output-form information the deployment's human-review flagging workflow requires. The deployment's binary 'flag for human review' decision derives from continuous model outputs, but BLURB does not characterize model behavior at decision boundaries or report calibration. The form match is therefore partial: representational forms align, but evaluation forms (calibration, threshold sensitivity) do not.

DRBENCHMARK evidence	BLURB evidence
[Q49] 'we have integrated all the training details, including hyperparameters, in Appendix A.' (p.5)	[Q76] 'we use the same train/dev/test split in Peng et al. and use Pearson correlation for evaluation.' (p.9)
[Q51] 'The reported results are obtained by averaging the scores from four separate runs ... statistical significance computed using Student's t-test.' (p.5)	[Q83] 'we instead adopt the original dataset and report micro F1 across the ten cancer hallmarks.' (p.9)
[Q52] 'we chose the SeqEval metric in conjunction with the IOB2 format and the training of all the models to predict only the label on the first token of each word' (p.6)	[Q100] 'We use cross-entropy loss for classification tasks and mean-square error for regression tasks.' (p.10)

[Q53] 'It provides a tokenizer-agnostic evaluation and mitigates any correlation between models' performances and the tokenization process.' (p.6)	[Q150] 'The BLURB score is the macro average of average test results for each of the six tasks' (p.14)
[Q54] 'For STS tasks, the models' performance was assessed using two metrics: (1) the Spearman correlation, and (2) the mean relative solution distance accuracy (EDRM)' (p.6)	[Q176] 'Comparison of entity-level F1 for biomedical named entity recognition (NER) using different tagging schemes' (p.19)
[Q92] 'although the benchmark is easily reproducible and customizable, it required a substantial amount of computational power to execute all runs.' (p.9)	
[Q93] 'approximately 2,500 hours on V100 GPUs' (p.9)	
[Q100] 'we locally save the best model based on its validation metric.' (p.13)	